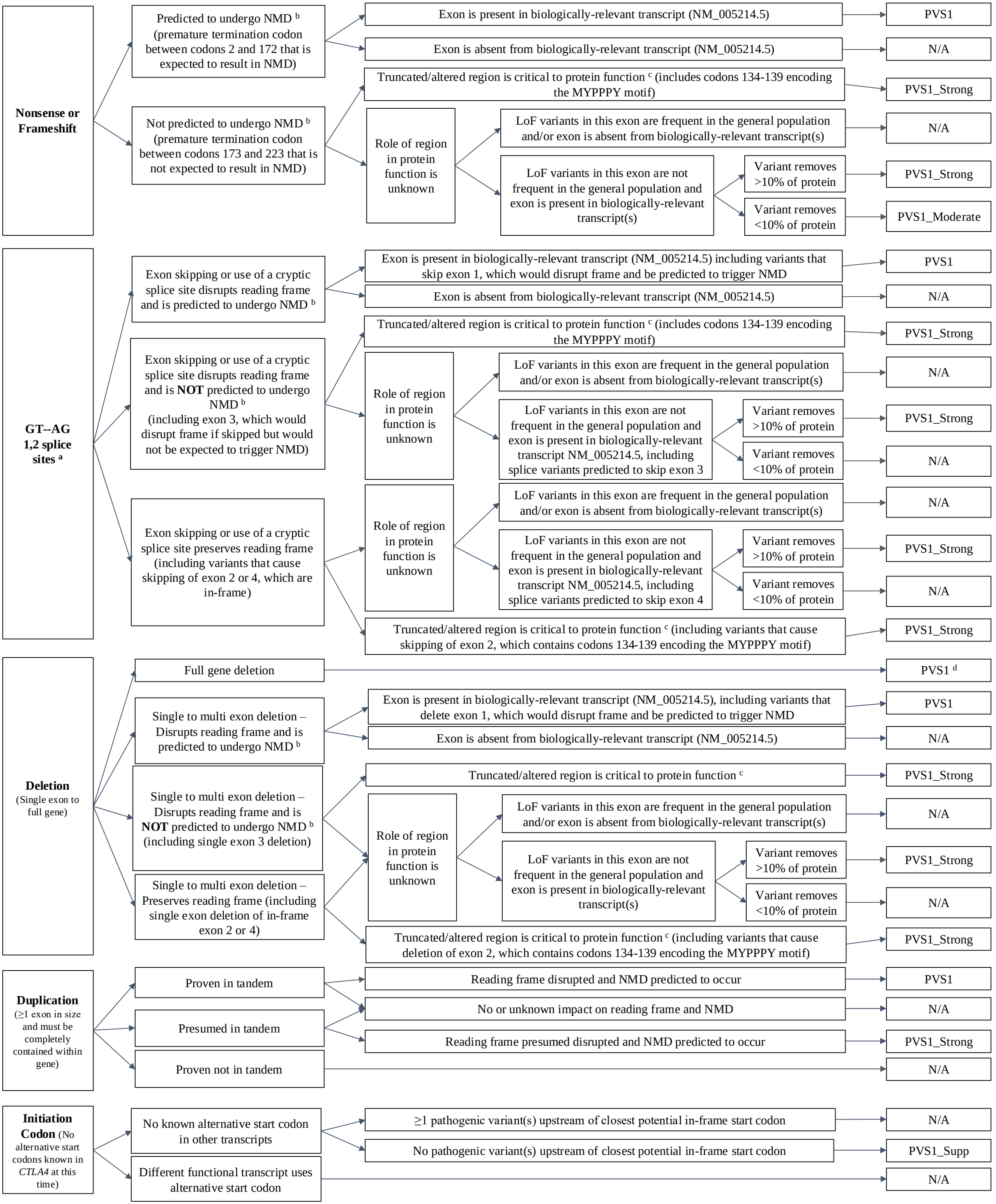

Nonsense or Frameshift
Predicted to undergo NMD [b] (premature termination codon between codons 2 and 172 that is expected to result in NMD)
Exon is present in biologically-relevant transcript (NM_005214.5)
PVS1
Exon is absent from biologically-relevant transcript (NM_005214.5)
N/A
Not predicted to undergo NMD [b] (premature termination codon between codons 173 and 223 that is not expected to result in NMD)
Truncated/altered region is critical to protein function [c] (includes codons 134-139 encoding the MYPPPY motif)
PVS1_Strong
Role of region in protein function is unknown
LoF variants in this exon are frequent in the general population and/or exon is absent from biologically-relevant transcript(s)
N/A
LoF variants in this exon are not frequent in the general population and exon is present in biologically-relevant transcript(s)
Variant removes >10% of protein
PVS1_Strong
Variant removes <10% of protein
PVS1_Moderate
GT--AG 1,2 splice sites [a]
Exon skipping or use of a cryptic splice site disrupts reading frame and is predicted to undergo NMD [b]
Exon is present in biologically-relevant transcript (NM_005214.5) including variants that skip exon 1, which would disrupt frame and be predicted to trigger NMD
PVS1
Exon is absent from biologically-relevant transcript (NM_005214.5)
N/A
Exon skipping or use of a cryptic splice site disrupts reading frame and is **NOT** predicted to undergo NMD [b] (including exon 3, which would disrupt frame if skipped but would not be expected to trigger NMD)
Truncated/altered region is critical to protein function [c] (includes codons 134-139 encoding the MYPPPY motif)
PVS1_Strong
Role of region in protein function is unknown
LoF variants in this exon are frequent in the general population and/or exon is absent from biologically-relevant transcript(s)
N/A
LoF variants in this exon are not frequent in the general population and exon is present in biologically-relevant transcript NM_005214.5, including splice variants predicted to skip exon 3
Variant removes >10% of protein
PVS1_Strong
Variant removes <10% of protein
N/A
Exon skipping or use of a cryptic splice site preserves reading frame (including variants that cause skipping of exon 2 or 4, which are in-frame)
Role of region in protein function is unknown
LoF variants in this exon are frequent in the general population and/or exon is absent from biologically-relevant transcript(s)
N/A
LoF variants in this exon are not frequent in the general population and exon is present in biologically-relevant transcript NM_005214.5, including splice variants predicted to skip exon 4
Variant removes >10% of protein
PVS1_Strong
Variant removes <10% of protein
N/A
Truncated/altered region is critical to protein function [c] (including variants that cause skipping of exon 2, which contains codons 134-139 encoding the MYPPPY motif)
PVS1_Strong
Deletion (Single exon to full gene)
Full gene deletion
PVS1 [d]
Single to multi exon deletion – Disrupts reading frame and is predicted to undergo NMD [b]
Exon is present in biologically-relevant transcript (NM_005214.5), including variants that delete exon 1, which would disrupt frame and be predicted to trigger NMD
PVS1
Exon is absent from biologically-relevant transcript (NM_005214.5)
N/A
Single to multi exon deletion – Disrupts reading frame and is **NOT** predicted to undergo NMD [b] (including single exon 3 deletion)
Truncated/altered region is critical to protein function [c]
PVS1_Strong
Role of region in protein function is unknown
LoF variants in this exon are frequent in the general population and/or exon is absent from biologically-relevant transcript(s)
N/A
LoF variants in this exon are not frequent in the general population and exon is present in biologically-relevant transcript(s)
Variant removes >10% of protein
PVS1_Strong
Variant removes <10% of protein
N/A
Single to multi exon deletion – Preserves reading frame (including single exon deletion of in-frame exon 2 or 4)
Truncated/altered region is critical to protein function [c] (including variants that cause deletion of exon 2, which contains codons 134-139 encoding the MYPPPY motif)
PVS1_Strong
Duplication (≥1 exon in size and must be completely contained within gene)
Proven in tandem
Reading frame disrupted and NMD predicted to occur
PVS1
Presumed in tandem
No or unknown impact on reading frame and NMD
N/A
Reading frame presumed disrupted and NMD predicted to occur
PVS1_Strong
Proven not in tandem
N/A
Initiation Codon (No alternative start codons known in *CTLA4* at this time)
No known alternative start codon in other transcripts
≥1 pathogenic variant(s) upstream of closest potential in-frame start codon
N/A
No pathogenic variant(s) upstream of closest potential in-frame start codon
PVS1_Supp
Different functional transcript uses alternative start codon
N/A